



Mumbai Quick Commerce Profitability & Operations Optimization Engine

Data-driven analysis of Blinkit and Zepto operations in Mumbai

Project Overview

Analysis of **19,826 Mumbai orders** across Blinkit and Zepto, built on a 1M+ record Q-commerce dataset with 12 engineered features, including dark store mapping, profit proxies, peak-hour flags, and churn risk scoring.

₹1.23Cr

Total GMV

₹24.5L

Total Profit

19.94%

Profit Margin

13.16m

Avg Delivery Time

70.66%

On-Time Rate

19,826

Total Orders

Tools: Python (pandas, seaborn, scikit-learn, Prophet) · SQL · Power BI

Key Findings at a Glance

Discount Margin Bleed

Excessive discounts are reducing margin on high-value orders in Personal Care and Snacks.

Andheri Overload

DS_001 (Andheri) handles **15% of all orders** but gets no additional rider support at peak hours.

Category Profit Gap

Personal Care generates **₹6.57L** in profit versus Dairy's ₹2.27L on nearly identical GMV.

Radius Loss Threshold

Orders beyond **4.5 km** are structurally loss-making; each additional kilometer adds to the loss.

Discount Churn Paradox

Low-rated customers still churn at **68%** after receiving a discount, so discounting is not the right retention lever.

Midday Dead Zone

Rider utilization falls to **28%** between 3 and 6 pm daily, creating a recurring infrastructure drain.

Category Profitability Gap

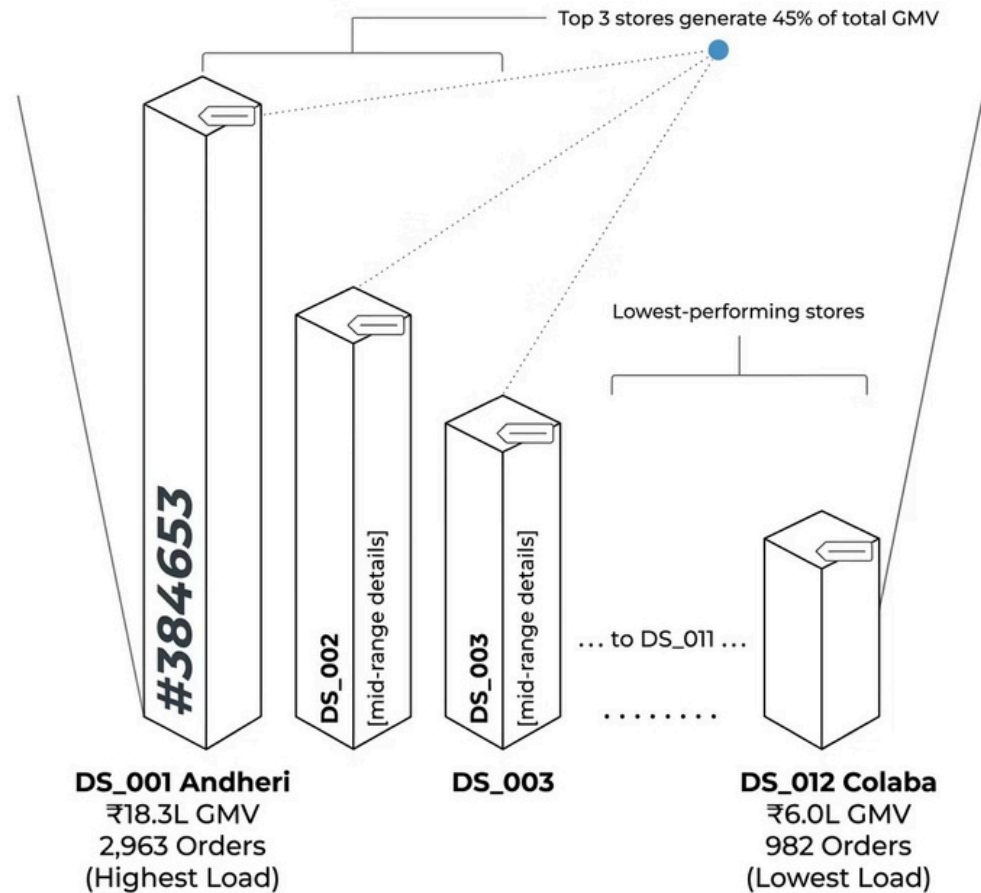
Category	GMV	Profit	Margin
Personal Care	₹ 1. 80M	₹6.57L	~36%
Household	₹ 1.75M	₹5.20L	~30%
Beverages	₹ 1.78M	₹3.35L	~19%
Dairy	₹ 1. 81M	₹2.27L	~13%
Fruits & Veg	₹ 1.70M	₹0.96L	~6%

Key Insight

GMV is nearly flat across categories, from ₹1.70M to ₹1.81M. Profit is not.

Dairy drives volume but contributes limited profit. Personal Care earns about 6× more profit per rupee of GMV than Fruits & Veg.

Dark Store Performance



Allocation issue

The **top 3 dark stores** (DS_001, DS_002, DS_003) generate **45% of total GMV**, but they are using the same rider pool as lower-demand locations.

⚠ Same rider pool. 3× demand difference. No reallocation logic.

Andheri – DS_001

₹18.3L GMV · 2,963 orders · **Highest demand**

Colaba – DS_012

₹6.0L GMV · 982 orders · **Lowest demand**

6 Strategic Recommendations

- P0

1

Cap Discounts on High-Value Orders

Cap at 15% for Personal Care & Snacks; remove discounts for orders >₹800. **Impact: ₹2.1Cr/year recovered.**
- P0

2

Enforce 4 km Delivery Hard Cap

Orders beyond this radius are unprofitable. **Impact: ₹21.6L/month saved.**
- P1

3

Dynamic Rider Reallocation

Shift riders to Andheri/Bandra from 7–10 pm. **Impact: churn reduced for 8K users/month.**
- P1

4

Replace Discount-as-Apology

Use a priority queue for rated ≤ 2 customers instead of discounts. **Impact: ₹43L/year LTV retained.**
- P2

5

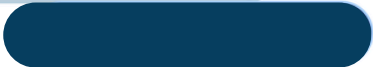
Homepage + Bundle Push

Promote Snacks and Personal Care bundles. **Impact: ₹1.8Cr GMV uplift.**
- P2

6

Scheduled Slots 3–6 pm

Offer a 10% discount to shift demand to off-peak delivery. **Impact: ₹34L/quarter infra savings.**



Total Addressable Impact

Implementing all 6 recommendations
could unlock ₹4–5Cr in annual value

₹2.1Cr

Discount Optimization

Recovered annually by capping and removing non-performing discounts

₹2.6Cr

Radius Savings

Annualized savings from enforcing the 4 km delivery cap

₹43L

Churn LTV Retained

Annual lifetime value preserved through service-first retention

₹1.8Cr

Category GMV Uplift

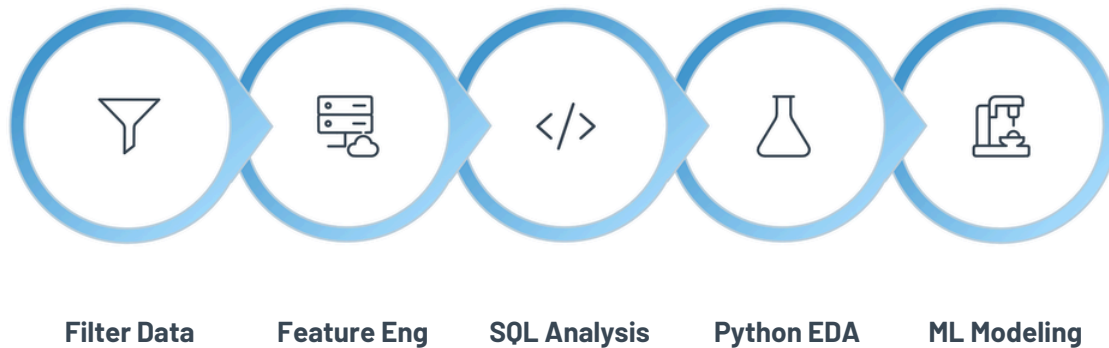
From homepage and bundle promotion of high-margin categories

₹1.36Cr

Infra Savings

Annualized savings from scheduled off-peak delivery slot adoption

Methodology



Analytical Stack

Python

pandas · seaborn ·
scikit-learn · Prophet

SQL

12 structured
business queries
across key
dimensions

Power BI

5-page interactive dashboard with DAX & what-if
scenarios

The Problem Is Efficiency, Not Revenue

Mumbai's Q-commerce profitability issue is not demand. It is operational efficiency and discount discipline. The datapoints to specific fixes, and the value from implementation is measurable.

19,826 Orders Analyzed

Across Blinkit and Zepto, Mumbai operations, 2023–2024

6 Actionable Levers

Ranked by impact, feasibility, and time to value

₹4–5Cr Unlockable

Estimated annual value from full implementation

